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Lanthanide-doped $Y_3Ga_5O_{12}$ garnets for nanoheating and nanothermometry in the first biological window

A.D. Lozano-Gorrín[a], U.R. Rodríguez-Mendoza[b], V. Venkatramu[c], V. Monteseguro[d], M.A. Hernández-Rodríguez[b], I.R. Martín[b], V. Lavín[b,*]

[a] Departamento de Química Inorgánica, and IMN, Universidad de La Laguna, Apdo. 456, 38200, San Cristóbal de La Laguna, Santa Cruz de Tenerife, Spain
[b] Departamento de Física, MALTA Consolider Team, IUdEA, and IMN, Universidad de La Laguna, Apdo. 456, 38200, San Cristóbal de La Laguna, Santa Cruz de Tenerife, Spain
[c] Department of Physics, Yogi Vemana University, Kadapa, 516 005, India
[d] Departamento de Física Aplicada - ICMUV, and MALTA Consolider Team, Edificio de Investigación, Universitat de València, C/Doctor Moliner 50, 46100, Burjassot, Valencia, Spain

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ABSTRACT

Absorption and luminescence spectra in the first biological window of Nd^{3+} single-doped and Er^{3+}-Yb^{3+} co-doped $Y_3Ga_5O_{12}$ nano-garnets have been studied to evaluate their potential use as simultaneous optical nano-heaters and nanothermometers in biomedicine. Nd^{3+}-doped nano-garnets uses the 808 nm laser radiation, resonant with the largest absorption peak of the $^4I_{9/2} \rightarrow {}^4F_{5/2}$ transition, for both heating the nanoparticle and populating the $^4F_{3/2}$ emitting level. Changes in the relative intensities of different emission peaks between Stark levels of the $^4F_{3/2}$ ($R_{1,2}$)$\rightarrow$$^4I_{9/2}$ ($Z_{1\text{-}5}$) transition can be directly related to the temperature of the nano-garnet. On the other hand, the Yb^{3+}/Er^{3+} combination takes advantage of the large absorption cross-section of the Yb^{3+} ions at around 920 nm to heat the sample, while triggering the Yb^{3+}-to-Er^{3+} upconversion energy transfer processes that populate the thermally coupled $^2H_{11/2}$,$^4S_{3/2}$ emitting levels of Er^{3+} ions, whose relative intensity changes with temperature can be calibrated. Accurate spatially- and temperature-controlled optically stimulated heating of these nano-garnets from room temperature up to 75 °C within the first biological windows is proved.

1. Introduction

Efficient luminescent nanostructures allowing, with a single laser beam, simultaneous optically-activated temperature sensing and heating, as well as imaging, for *in vivo* thermal treatments of subtissue tumors has been a hot topic in biomedicine in the last years [1]. Metallic nanoparticles, quantum dots, carbon nanostructures, porous silicon, organic nanoparticles and lanthanide-doped nanocrystals have been extensively studied for photothermal therapies [1–3]. The hyperthermia treatment, in which the cancer cells are heated to the cytotoxic level (43–45 °C), needs a careful and accurate thermal sensing for *real-time* control of the absorbing energy of the laser radiation, minimizing the collateral damage that could arise from excessive heating [1–7].

The first difficulty arises from the optical radiation that should be used, since the ultraviolet and the high energy part of the visible light cannot penetrate efficiently into human tissues because of the combination of light scattering, due to tissue inhomogeneities, and light absorption, caused by different compounds present in real tissues, such as hemoglobin and water [8]. In order to overcome this limitation, the luminescent nanoparticles used in biophotonic applications should be excited and should emit in the biological windows (BWs) of the human tissues, i.e. in those energy ranges in which the tissue is more transparent to optical radiation, assuring greater penetration and lower damage [9]. The first biological window (1-BW) ranges from ~650 to ~970 nm, in which the absorption of water and other compounds, such as hemoglobin, are relatively weak. The 2-BW, located entirely in the near-infrared (NIR) range, extends from ~1000 to ~1400 nm, limited by two strong absorption bands of water, whereas the 3-BW extends from ~1500 to ~1700 nm also in the NIR range. For example, carbon nanotubes have been used for *in vivo* imaging using the 1-BW [10] and lanthanide-doped nanoparticles have been used for *real-time* infrared *in vivo* imaging within the 2-BW [11].

Conventional infrared thermometry based on the black body radiation only provides accurate temperature at the skin's surface. Nowadays, the emergence of nanotechnology has opened new lines of research for the use of the luminescent nanoparticles for subtissue thermal sensing [9,12–14]. Nanothermometry is especially important in

* Corresponding author.
E-mail address: vlavin@ull.edu.es (V. Lavín).

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biomedicine and can help extracting knowledge of the local dynamics and performance of the majority of biological microorganism (cell, bacteria, …) that are strongly determined by temperature, and whose sizes are larger than the nanoparticles. Optical nanothermometers are calibrated using different techniques (changes in lifetime, polarization, bandwidth, etc.), although the most widely used is that in which the relative intensities of the emissions from two closed-energy, thermalized levels is related to temperature [15–19]. The luminescences of quantum dots, colloidal and lanthanide (Pr^{3+}, Nd^{3+}, Er^{3+} or Tm^{3+}) doped nanocrystals have been calibrated as local temperature probes [12–14,20]. Moreover, in biomedicine applications the strategies for improving the temperature detection at the nanometer scale in the first biological window has involved the synthesis of sophisticated core-shell nanoarchitectures in combination with non-linear upconversion processes using different lanthanide ions [21,22].

After laser excitation, heat is induced by the generation of phonon modes in the vibrating nanostructure, whose energy is subsequently transferred to the surrounding medium. The thermal sensitivity plays an important role in the careful control of the spatial location and heat transfer rate to the cell, which is directly related to the nanometer size of the particle and its large surface-to-volume ratio. Garnet nanostructures show good biologic compatibility and relatively small cytotoxicity, compared to quantum dots, and they do not need functionalization to enter in the cells. In addition, they show high transparency from the UV to the mid-IR range, good chemical stability, high thermal conductivity, and high energy phonons to generate heat [20,23].

For the design and synthesis of multifunctional nanoparticles capable of simultaneous heating and thermal sensing, doping with lanthanide trivalent ions has the advantage of having multiple absorption and emission bands in the BWs, with multiple combinations of NIR excitations and NIR detections [1]. The first attempt to optically stimulate heating was done by Bednarkiewicz *et al.* [4] using Nd^{3+}-doped $NaYF_4$ nanoparticles and, more recently, Nd^{3+}-doped LaF_3 nanoparticles have been used for simultaneous thermal therapy, as nanoheater and nanothermometer, and *in vivo* imaging using the 1- and 2-BWs [7].

In this work we analyze the absorption and luminescence properties of Nd^{3+} single-doped and Yb^{3+}/Er^{3+} co-doped lanthanides in $Y_3Ga_5O_{12}$ nano-garnets as simultaneous heaters and thermometers in the wavelength range of the 1-BW from ~780 to ~920 nm, and in the biological temperature range, from room temperature up to 70 °C.

2. Synthesis and experimental techniques

Nanocrystalline yttrium gallium garnets of composition $Y_{3(1-x)}Ln_{3x}Ga_5O_{12}$ (YGG), with x = 0.01, and 0.10 and Ln = Nd, Er, and Yb, were synthesized by the citrate sol-gel method in air [20,23–25]. These moderate doping concentrations have allowed making a proof-of-concept for the real ability of using the YGG nanogarnets as optically-controlled multifunctional nanoparticles for simultaneous heating and thermal sensing purposes, using laser excitation and spontaneous photon emission within the 1-BW, although higher concentrations of lanthanide ions (Nd^{3+} and Yb^{3+} in this work) are expected to benefit the light-to-heat conversion [4,6].

The powder was obtained as agglomerated nanocrystals with different shapes and sizes ranging from 40 to 60 nm (see Fig. 1) [23,24]. In addition, they show low cytotoxicity, at least with concentrations up to 120 μg/mL, and do not need any functionalization to enter the cell [26]. X-ray powder diffraction (XRPD) patterns were measured on a diffractometer PANalytical X'Pert Pro using $CuK_{\alpha 1}$ radiation. The fitted XRPD pattern for YGG is shown in Fig. 1. All the reflections are well indexed to a single cubic phase with space group *Ia*-3d (No. 230, Z = 8), where no impurity phase has been detected. The corresponding unit cell parameters have been obtained by fitting the profiles of the nanogarnets by the Rietveld method [27] using FullProf program [28]. According to the reliability factors (χ^2 = 8.8; R_p = 12.8; R_{wp} = 11.7;

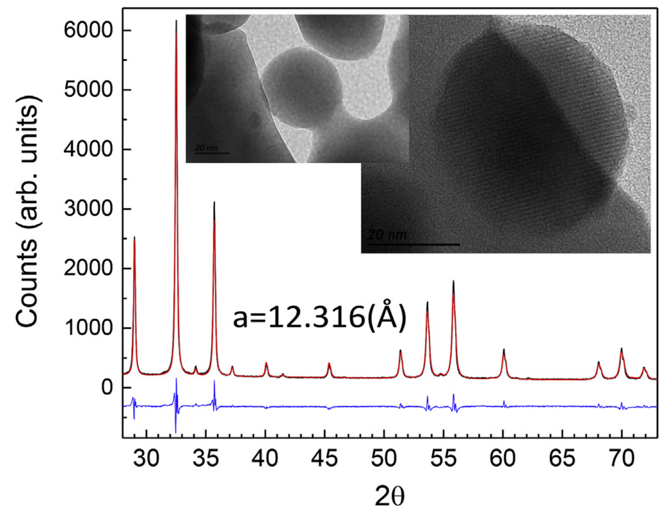

Fig. 1. X-ray diffraction pattern of YGG nano-garnets (black). Rietveld refinements including the differences (blue) between calculated (red) and observed (black) patterns are also shown. HRTEM micrograph (20 nm scale) of YGG nano-garnets. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

R_{exp} = 3.94), quite goodness of fitting parameters has been reached. The garnet structure can be described as a network of GaO_6 octahedra and GaO_4 tetrahedra linked by shared oxygen ions at the corners. The polyhedra are arranged in chains along the three crystallographic axes and form dodecahedral holes, with D_2 local point symmetry, which are occupied by the Y^{3+} and Ln^{3+} ions. The structural, vibrational and optical properties of these nano-garnets are quite similar to those of the bulk garnet crystal [20,23–25].

Diffuse reflectance spectra of the powder samples have been measured with a spectrophotometer (Agilent Cary 5000) equipped with an integrating sphere. The Stokes and upconverted emission spectra were measured with a 0.3 m focal length spectrograph (Andor Shamrock 303i) equipped with a Peltier cooled silicon CCD camera (Andor Newton) by exciting the lanthanides with a tunable cw Ti: sapphire laser (Spectra Physics 3900 S) pumped by a 532 nm diode laser (Spectra Physics Millenia Prime Laser 15s JSPG). For temperature calibration, nanoparticles were located at the center of a tubular furnace (± 0.7 °C resolution) and measurements were taken following a 180° configuration [17–19]. During laser heating, a period of 2 min where taken to achieve the steady-state temperature regime before measuring. All spectra were corrected from instrument response.

3. Results

The optical properties of the lanthanide ions depend on the local structure of these ions in the nano-garnets, since it rules the fine structure splitting of the $^{2S+1}L_J$ multiplets and the intra-configurational *4f-4f* electric-dipole transitions probabilities [29]. Lanthanide ions will occupy the 8-coordinated orthorhombically distorted D_2 dodecahedral sites replacing the Y^{3+} ions without charge compensation, which completely breaks the free-Nd^{3+}, -Er^{3+} and -Yb^{3+} ions degeneracies and giving rise to (2 J + 1)/2 Stark, or crystal-field, levels [20,23–25] (see Fig. 2).

The studies of the Nd^{3+} and Yb^{3+}/Er^{3+} in the YGG nano-garnets as simultaneous heaters and thermometers in the 780–920 nm NIR range of the 1-BW have followed two main steps: firstly, an accurate optical calibration of the temperature using the fluorescence intensity ratio (FIR) technique [9,17–19]; and secondly, the correlation between the laser pump density and the local temperature of the area irradiated by the laser [4,7].

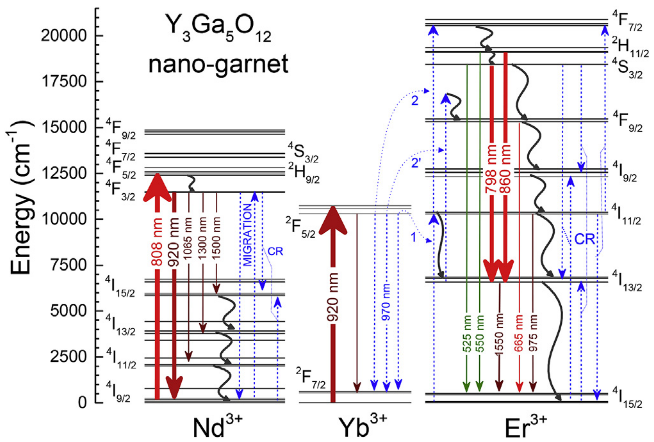

Fig. 2. Partial energy level diagrams of Nd^{3+}, Yb^{3+} and Er^{3+} ions in $Y_3Ga_5O_{12}$ (YGG) nanogarnets. Electronic absorption and emission radiative transitions (solid colored arrows), energy transfer non-radiative de-excitations (blue dashed arrows) and multiphonon relaxations (wavy dark gray arrows) are also indicated. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.1. Nd^{3+}-doped $Y_3Ga_5O_{12}$ nano-garnet

Nd^{3+} ion shows relatively strong absorption and high quantum yield emission bands in the visible and NIR ranges of the 1-BW [24], although Fig. 3 only shows the $^4I_{9/2}\rightarrow{}^4F_{5/2},{}^2H_{9/2}$ absorption band in the 780–830 nm range and the $^4F_{3/2}\rightarrow{}^4I_{9/2}$ emission band in the 860–940 nm one that will be used for the simultaneous heating and temperature sensing effects [6]. As it can be observed in the diffuse reflectance spectrum given in Fig. 3, the 808 nm laser photon energy is resonant with the maximum absorption peak of the $^4I_{9/2}\rightarrow{}^4F_{5/2},{}^2H_{9/2}$ intra-configurational *4f* ^{3}-*4f*^{3} electronic transitions, which can be tentatively assigned to the transition between the lowest Stark levels of the $^4I_{9/2}$ ground state and the $^4F_{5/2}$ excited multiplet. After the laser excitation at 808 nm, the Nd^{3+} ions immediately de-excite interacting with the vibrating garnet network, generating at least two phonons to bridge the energy gap of 890 cm^{-1} and populate the two thermalized emitting levels of the $^4F_{3/2}$ multiplet. The emitting multiplet has an experimental lifetime of ~280 μs [24], typical of Nd^{3+} ions in oxides and a few orders of magnitude longer than the nanosecond fluorescence lifetimes found in organic fluophores [4], with enormous interest in bioimaging.

Temperature produces a slight energy shift of the emission peaks in the $^4F_{3/2}\rightarrow{}^4I_{9/2}$ transition [5], as well as more important changes in some of their relative intensities (see Fig. 3). The FIR applied to Nd^{3+} ions relates these relative changes to emission from the two closed energy (~35 cm^{-1}), thermalized R_1 and R_2 Stark levels of the $^4F_{3/2}$ emitting multiplet to temperature. In fact, the emission band associated with the $^4F_{3/2}\rightarrow{}^4I_{9/2}$ transition is due to the overlapping of emissions from the R_1 and R_2 emitting levels to the Z_1-Z_5 five Stark levels of the ground state [24]. Since the populations of the low-energy R_1 and high-energy R_2 Stark levels obeys the Boltzmann temperature distribution, the emission peaks from R_2 level increase with temperature, while those departing from the R_1 level decrease [5–8].

As it can be seen in Fig. 3, the largest changes when temperature increases from RT up to 73.5 °C are observed for those peaks in the 879.75–886.7 nm range (labelled as IR2), whereas only slight energy shift and intensity change are observed for the high energy peak in the 870–873.5 nm range (labelled as IR1) [7]. The latter peak can be tentatively associated with the $^4F_{3/2}$ (R_1)→$^4I_{9/2}$ (Z_1) transition, although a small overlap with the R_2→Z_1 emission cannot be disregarded. On the other side, those peaks in the IR2 range are due to the overlapped emissions of the $^4F_{3/2}$ (R_2)→$^4I_{9/2}$ (Z_3) and $^4F_{3/2}$ (R_1)→$^4I_{9/2}$ ($Z_{2,3}$) transitions, and their relative changes can be understood as a due to

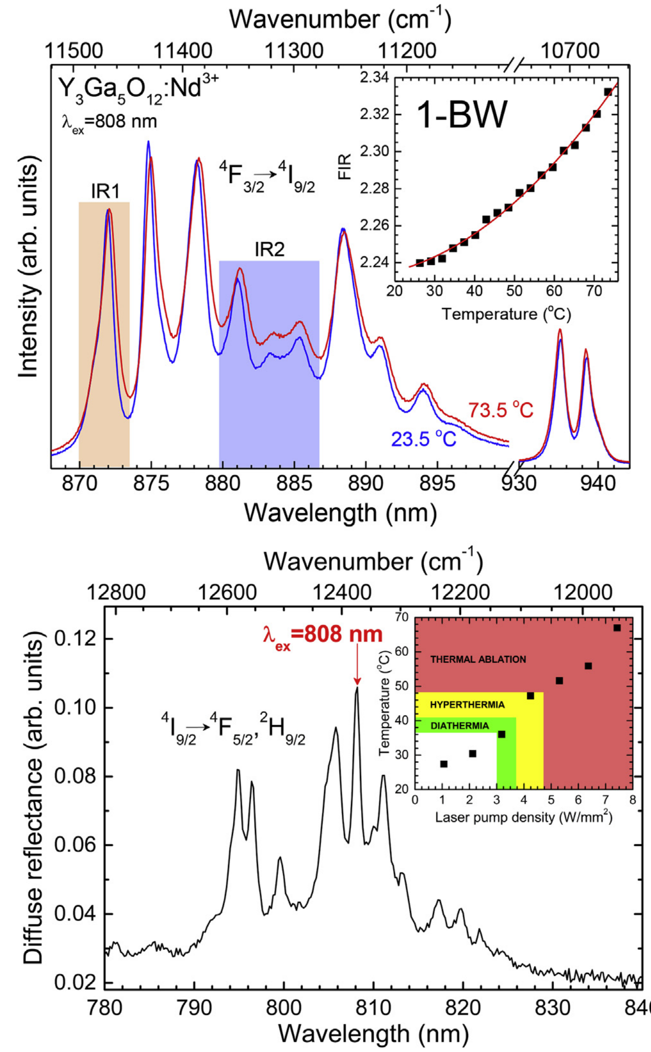

Fig. 3. (Top) Nd^{3+} emission spectrum associated with the $^4F_{3/2}\rightarrow{}^4I_{9/2}$ transition in $Y_3Ga_5O_{12}$ nano-garnets at two different temperatures within the thermal biological range. Inset shows the IR2/IR1 FIR as a function of temperature and the fitting to a Boltzmann distribution (see text). (Bottom) Nd^{3+} absorption spectrum associated with the $^4I_{9/2}\rightarrow{}^4F_{5/2},{}^2H_{9/2}$ transitions in $Y_3Ga_5O_{12}$ nano-garnets at room temperature. Inset shows the temperature reached by the nano-garnet as a function of the 808 nm laser pump power density. Colors indicate the different regimes for thermal therapies in biomedicine. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

populations changes of the R_1 and R_2 emitting levels. The intensity ratio of the peaks in the IR2 and IR1 emitting ranges, measured using appropriate bandpass filters, shows a non-linear variation with temperature in the biological thermal range (see Fig. 3), that can be fitted to the Boltzmann distribution, giving rise to a thermal sensitivity S (=1/FIR· d (FIR)/dT) of 1.3×10^{-3} °C^{-1} at 77 K, with temperature resolution of ± 2 °C. These results are only slightly smaller than those found in YAG ($Y_3Al_5O_{12}$) or $NaYF_4$ nanoparticles, although different to the almost linear variation found for Nd^{3+}-doped LaF_3 nanoparticles, which show a higher thermal sensitivity of ~2.5×10^{-3} °C^{-1} [7].

Once the ratio of emissions of the Nd^{3+} ions is calibrated, the possibility to manipulate, in a controlled way, the temperature of the nano-garnet depends only on its capacity to achieve a large laser-induced heat generation. For heating purposes, and as pointed out by Bednarkiewicz *et al.* [4], the 808 nm laser radiation was chosen for different reasons: i) the absorption bands of the 1-BW shows a maximum at this wavelength, ii) there are commercially available laser diodes for excitation, and iii) the tissue 1-BW optical transmission shows low absorption coefficient for a phantom tissue ($< 6\,cm^{-1}$), which mimic that of an "*in vitro*" human skin tissue, hemoglobin ($\sim 4\,cm^{-1}$) and water ($\sim 0.02\,cm^{-1}$), allowing the light to penetrate a few centimeters (larger compared to UV or visible radiations), only perturbed by light scattering processes [4,5].

The nanoheater efficiency takes advantage of the IR-active 840 cm^{-1} and Raman-active 755 cm^{-1} high energy phonon modes, as well as other intense phonons at 354, 528 and 600 cm^{-1} found in YGG nano-garnets [24]. As pointed out by Rocha *et al.* [6], heat is generated as a consequence of both the increase of the absorbed laser pump power, which apart from the small intrinsic absorption of the network also excites a large number of Nd^{3+} ions to high energy levels, and the decrease of its luminescence quantum yield, which increases the non-radiative de-excitation probabilities that produce the generation of phonons in the garnet network, which finally transfer this heat to the surrounding medium. Actually, heating the nano-garnet network and, hence, the surrounding area means that part of the 808 nm laser photons absorbed have been invested in the generation of phonons due to the interaction of the *4f* electrons of the Nd^{3+} ion with the vibrating nano-garnet network. There are two main ways to generate phonons: Once one Nd^{3+} ion is promoted to the $^4F_{5/2}$ level, the multiphonon processes start to act while the Nd^{3+} ion loses the energy to reach the $R_{1,2}$ emitting levels. As already commented, from the metastable $^4F_{3/2}$ metastable state, Nd^{3+} ions can decay radiatively (spontaneous emission of photons) to any of the lower energy ^4I_J (J = 9/2–15/2) multiplets, since the energy gap is so large that the probability to bridge it generating phonons is negligible. However, the energy gaps between the ^4I_J multiplets are small enough to allow the de-excitation of the Nd^{3+} ions via non-radiative multiphonon relaxations. The latter processes will be the main supplies of energy that will heat the nano-garnet doped with Nd^{3+} ions.

On the other side, there are cross-relaxation and migration energy transfer processes among Nd^{3+} ions, as well as from Nd^{3+} ions to impurities or trap, that give rise to another source of non-radiative de-excitation channels for the Nd^{3+} ions to return to the ground state (see Fig. 2). However, these processes will also populate the ^4I_J multiplets, while impurities and traps will interact directly with the network, both increasing the heating. The doping concentration used for heating is 1 mol% of Nd_2O_3, large enough to have a reasonable high absorption at 808 nm, while providing an intense luminescence and an experimental lifetime for the $^4F_{3/2}$ emitting level of $\tau \sim 280\,\mu s$, close enough to the intrinsic lifetime of $\tau_0 \sim 315\,\mu s$ expected when non-resonant cross-relaxation energy transfer processes are negligible. This gives a rough value for the ($^4F_{3/2}$,$^4I_{15/2} \rightarrow {}^4F_{3/2}$,$^4I_{15/2}$) channel's energy transfer probability, evaluated as $W_{ET} = (\tau)^{-1}$-$(\tau_0)^{-1}$, of $\sim 400\ s^{-1}$, indicating that the multiphonon processes are the main responsible for heating the garnet nanocrystalline structure.

Light-to-heat conversion has been obtained in the Nd^{3+}-doped YGG nano-garnet in the biological temperature range and it is shown as an inset in Fig. 3. From these results it is clear that a laser power density at least one order of magnitude larger than that used for gold nanostructures is necessary [30], although perfectly achievable by commercial laser diodes and sufficient to perform localized photo-thermo-therapies.

3.2. Er^{3+}-Yb^{3+} co-doped $Y_3Ga_5O_{12}$ nano-garnet

Yb^{3+} ion shows a very simple energy level scheme with only two multiplets, the $^2F_{7/2}$ ground state, with four Stark levels, and the $^2F_{5/2}$ excited level, with three Stark levels, separated by a large energy gap of $\sim 10{,}000\,cm^{-1}$ (see Fig. 2). It also shows a large and broad absorption band with a maximum at ~ 970 nm but, more interesting and as a especial characteristic of the garnet structure, there is a large absorption coefficient for the overlapped peaks in the 920–945 nm range, similar to that of the maximum, that open the possibility of obtaining high quantum efficiency as nanoheater in the 1-BW, far away from the absorption of human tissue.

Since the heating and temperature sensing wants to be performed using only a single laser radiation with an energy resonant with the $^2F_{7/2} \rightarrow {}^2F_{5/2}$ absorption band of the Yb^{3+} ions, it is necessary to find a particular laser excitation within the range of wavelengths that this band overlaps with the 1-BW. As Vetrone *et al.* [9], a 920 nm laser radiation has been chosen since: i) it shows an absorption coefficient large enough to promote a large number of Yb^{3+} ions, ii) the absorption coefficients of hemoglobin ($\sim 4\,cm^{-1}$) and water ($\sim 0.1\,cm^{-1}$) are still relatively low, and iii) it is still possible to trigger those mechanisms necessary to excite the Er^{3+} ions to the $^2H_{11/2}$ and $^4S_{3/2}$ thermalized emitting levels via Yb^{3+}-to-Er^{3+} upconversion energy transfer processes.

These upconversion processes have been previously studied for $Lu_2Ga_5O_{12}$ (LuGG) nano-garnet [25], but no important differences are expected for the YGG since their structural, vibrational and optical properties are very similar [24]. Under 920 nm laser excitation an intense green luminescence from Er^{3+} ions can be observed by the naked eyes, indicating that an efficient Yb^{3+}-to-Er^{3+} energy transfer upconversion processes are the main responsible for the Er^{3+}, since no resonant absorption of Er^{3+} is possible. The main processes that describe the NIR-to-visible upconversion are presented in Fig. 2: once Yb^{3+} ions populate the $^2F_{5/2}$ excited state by the resonant absorption of the laser radiation, one Yb^{3+} ion relaxes to the $^2F_{7/2}$ ground state and the energy of the transition is transferred to one Er^{3+} ion in the $^4I_{15/2}$ ground state to populate the $^4I_{11/2}$ intermediate state (see label "1" in Fig. 2). From this metastable level, the Er^{3+} can absorb another quantum of energy from a second Yb^{3+} ion to populate the $^4F_{7/2}$ state (see label "2" in Fig. 2), from which a phonon-assisted de-excitation relaxes the Er^{3+} ion to the thermalized levels. Part of the energy involved in the de-excitation from these levels is spontaneously emitted in form of green, red, and NIR photons. It is worth noting that only a small fraction of the Yb^{3+} population in the $^2F_{5/2}$ state participates in the upconversion processes, since the large majority of them relaxes to the ground state emitting in a wide range of wavelengths, i.e. from 920 to 1035 nm, with a maximum at ~ 970 nm. In addition, and complicating even more this picture, a small part of this photons can be also absorbed by Er^{3+} ions to promote to the $^4I_{11/2}$ metastable intermediate level.

The FIR technique has been successfully applied to different Er^{3+}-doped bulk materials as well as nanomaterials [9,17–20]. Although the green $^2H_{11/2}$,$^4S_{11/2} \rightarrow {}^4I_{15/2}$ emissions in the 520–550 nm range show a high thermal sensitivity for the YGG nano-garnet and other nanomaterials [9,20], for biomedicine applications is better to use the temperature–dependent emissions associated with the $^2H_{11/2}$,$^4S_{3/2} \rightarrow {}^4I_{13/2}$ transition, whose photons show energies in the 780–870 nm NIR range of the 1-BW. These are the emissions used to calibrate the temperature of the nano-garnet.

The $^2H_{11/2}$ multiplet is $\sim 800\,cm^{-1}$ higher in energy than the $^4S_{3/2}$ doublet one and both have a hyperfine structure of Stark levels that are reflected in their particular emission spectrum. In Fig. 4 the emission spectra at two different temperatures at the edges of the biological thermal range are presented, normalized to the area of the $^4S_{3/2} \rightarrow {}^4I_{13/2}$ band. The case for Er^{3+} ions is simpler than that for Nd^{3+} ions, since the FIR parameter can be easily calculated as a function of temperature as the ratio of intensities, $I_2({}^2H_{11/2} \rightarrow {}^4I_{13/2})/I_1({}^4S_{3/2} \rightarrow {}^4I_{13/2})$, and it can be fitted to a Boltzmann distribution, $I_2/I_1 \propto \exp(-\Delta E/kT)$, of the $^2H_{11/2}$ and $^4S_{3/2}$ thermalized levels, giving rise to a thermal sensitivity of the order of $\sim 8.0 \times 10^{-3}\,{}^\circ C^{-1}$ at 77 °C.

Light-to-heat conversion is shown in Fig. 3 for the Yb^{3+}/Er^{3+} co-doped YGG nano-garnet. The 920 nm laser radiation seems to be enough to heat the sample through different non-radiative processes that generate phonons by interactions with the vibrating garnet network, apart from its weak intrinsic absorption at this wavelength. As it can be seen in Fig. 2, the cascade type de-excitations of Er^{3+} ions starting from the $^4F_{7/2}$ multiplet to the ground state, via all the intermediate levels,

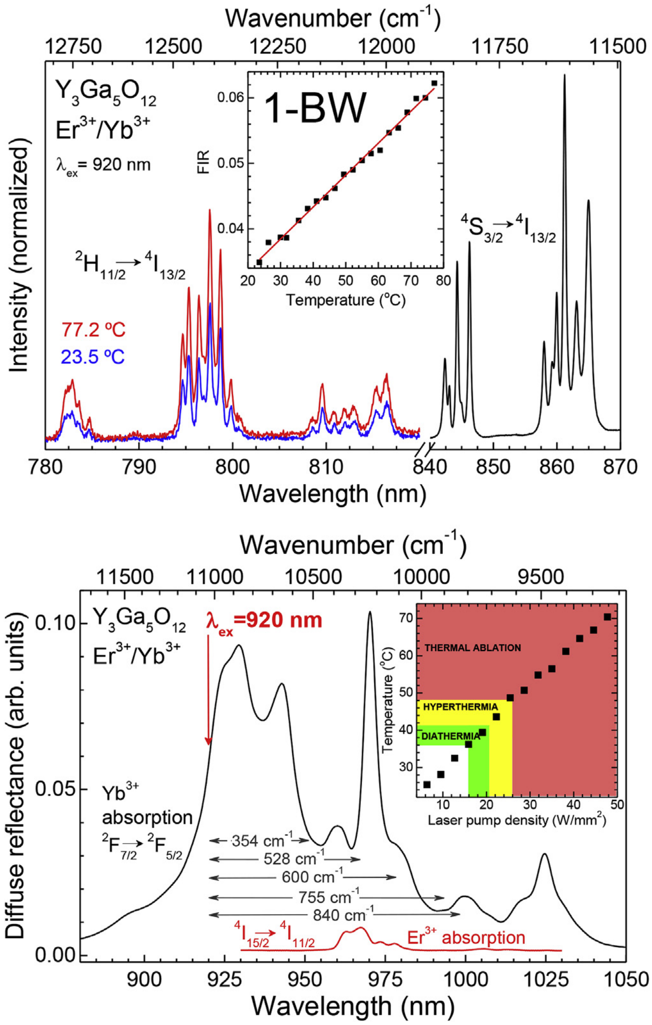

Fig. 4. (Top) Er^{3+} emission spectrum associated with the $^2H_{11/2},^4S_{3/2} \rightarrow {}^4I_{13/2}$ transitions in $Y_3Ga_5O_{12}$ nano-garnets at two different temperatures within the thermal biological range. Inset shows the $I_2(^2H_{11/2} \rightarrow {}^4I_{13/2})/I_1(^4S_{3/2} \rightarrow {}^4I_{13/2})$ fluorescence intensity ratio as a function of temperature and the fitting to a Boltzmann distribution (see text). (Bottom) Yb^{3+} and Er^{3+} absorption spectra associated with the $^2F_{7/2} \rightarrow {}^2F_{5/2}$ and $^4I_{15/2} \rightarrow {}^4I_{11/2}$ transitions, respectively, in $Y_3Ga_5O_{12}$ nano-garnets at room temperature. Energies of the most intense IR- and Raman-active phonon modes are also shown. Inset shows the temperature reached by the nano-garnet as a function of the 920 nm laser pump power density. Colors indicate the different regimes for thermal therapies in biomedicine. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

generate large number of phonons necessary for heating the nanostructure, including the phonon-assisted upconversion energy transfer (labelled as 2' in Fig. 2). However, the number of Er^{3+} ions in these processes are not large, and the efficiencies in generating phonons is limited, even if energy transfer cross-relaxation processes are involved (see some of the most interesting resonant CR channels in Fig. 2). On the other side, the contribution to the generation of phonons of the energy gap that exist between the 920 nm laser radiation and the Yb^{3+} emission and Er^{3+} absorption at around 970 nm, i.e. a gap of $\sim 560\ cm^{-1}$ that has to be bridged to trigger efficiently the upconversion processes, takes place in a wide range of energies that would involve all the phonon modes in the garnet, and seems to be the largest contribution. As a result, a laser power density at least one order of magnitude larger compared to Nd^{3+}- doped nano-garnets, and two orders compared with gold nanoparticles, is necessary for the concentrations of Yb^{3+} and Er^{3+} ions used in this work.

4. Conclusions

The potential use of Nd^{3+} single-doped and Yb^{3+}/Er^{3+} co-doped $Y_3Ga_5O_{12}$ nano-garnet to perform localized heating and temperature sensing thermo-therapies has been studied. The Nd^{3+} ion allows the simultaneous heating and thermometry by absorbing resonantly the 808 nm laser radiation energy, associated with the $^4I_{9/2} \rightarrow {}^2H_{9/2},^4F_{5/2}$ transitions, to subsequently heat the sample through the generation of phonons when interacting with the vibrational network of the garnet structure. At the same time, these non-radiative de-excitation processes also populates the $^4F_{3/2}$ emitting level, whose temperature-sensitive luminescence to the first excited $^4I_{11/2}$ multiplet in the near-infrared range will allow calibrating the temperature of the nano-garnet. On the other hand, the co-doped nano-garnet takes advantage of the large absorption of energy of the Yb^{3+} ions at 920 nm to heat the nanoparticles, while triggering the NIR-to-visible Yb^{3+}-to-Er^{3+} upconversion energy transfer processes that will populate the thermalized $^4S_{3/2},^2H_{11/2}$ emitting levels of Er^{3+} ion, whose emissions to the first excited $^4I_{13/2}$ are used to calibrate the temperature. All these excitation-detection processes take place in the 780–920 nm NIR range of the high transmitting 1-BW of the human tissue. In the case of Nd^{3+} ions, a laser power density at least one order of magnitude larger compared to gold nanoparticles is necessary, whereas for the concentration of Yb^{3+}/Er^{3+} combination of lanthanides at least two orders of magnitudes are necessary. These laser pump densities are perfectly achievable by commercial laser diodes and sufficient to perform localized photo-thermo-therapies.

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